

Clinical Risk Assessment Report: Patient 0

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 47.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.14x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.4%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.49 [Bottom <1%] (+7.4%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.62 [Top 25%] (+4.4%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.03 [Top 13%] (+3.5%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.72 [Avg (71th %ile)] (+3.3%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.62 [Top 1%] (+2.6%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.55 [Bottom 18%] (+2.4%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.29 [Bottom <1%] (+2.2%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.61 [Top 4%] (+1.8%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

